



**Gradient Descent. Convergence for  
quadratics; smooth convex case; PL case**

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# Gradient Descent

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The result of this method is

$$x_{k+1} = x_k - \alpha f'(x_k)$$

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Let's consider the following ODE, which is referred to as the Gradient Flow equation.

$$\frac{dx}{dt} = -f'(x(t)) \quad (\text{GF})$$

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(GF)

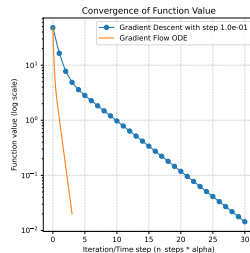
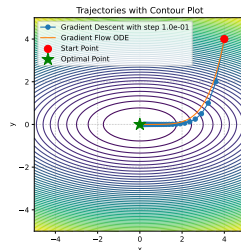
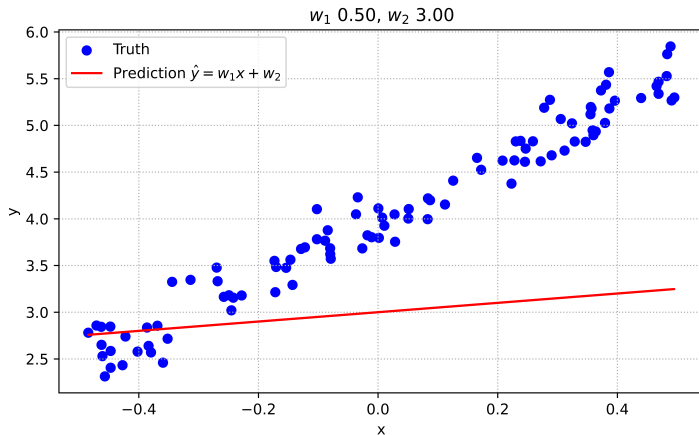
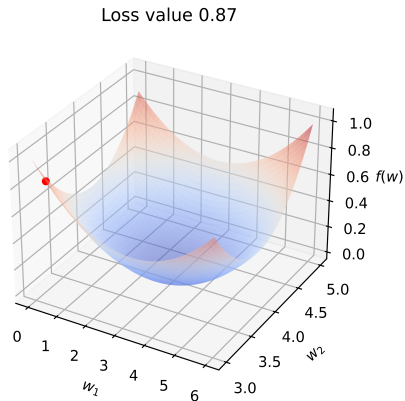


Figure 1: Gradient flow trajectory

# Convergence of Gradient Descent algorithm

Heavily depends on the choice of the learning rate  $\alpha$ :



## Exact line search aka steepest descent

$$\alpha_k = \arg \min_{\alpha \in \mathbb{R}^+} f(x_{k+1}) = \arg \min_{\alpha \in \mathbb{R}^+} f(x_k - \alpha \nabla f(x_k))$$

More theoretical than practical approach. It also allows you to analyze the convergence, but often exact line search can be difficult if the function calculation takes too long or costs a lot. An interesting theoretical property of this method is that each following iteration is orthogonal to the previous one:

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$$\nabla f(x_{k+1})^\top \nabla f(x_k) = 0$$

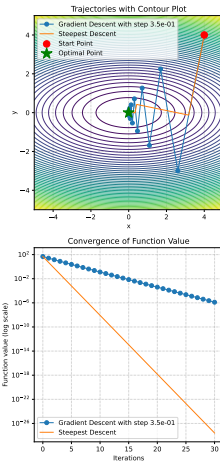


Figure 2: Steepest Descent

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## Strongly convex quadratics

## Coordinate shift

Consider the following quadratic optimization problem:

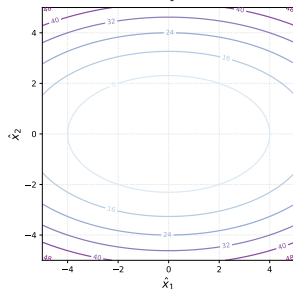
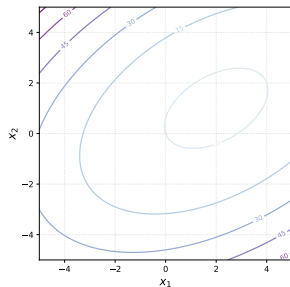
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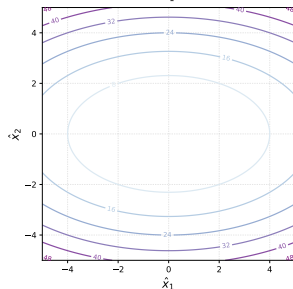
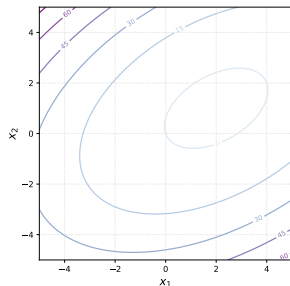


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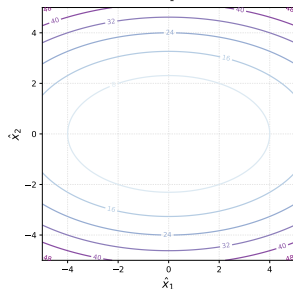
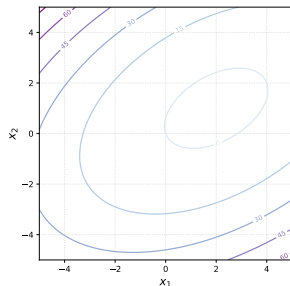


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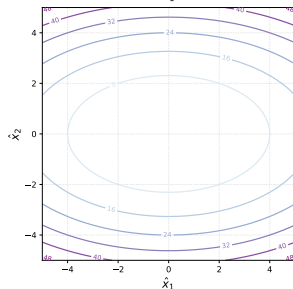
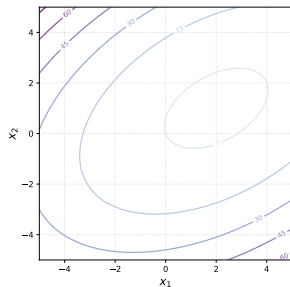
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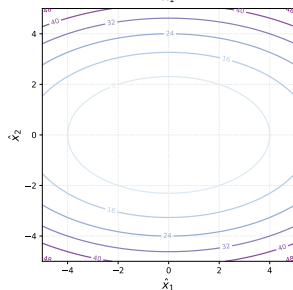
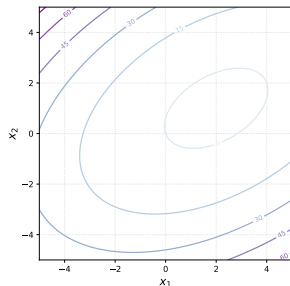
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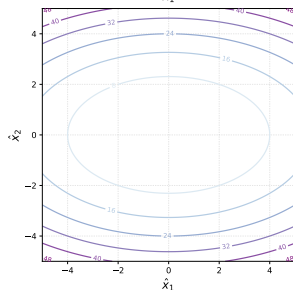
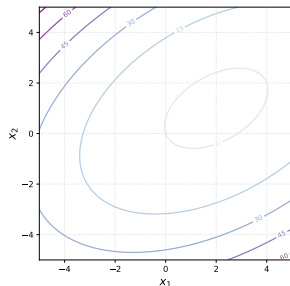
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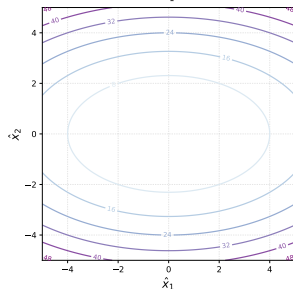
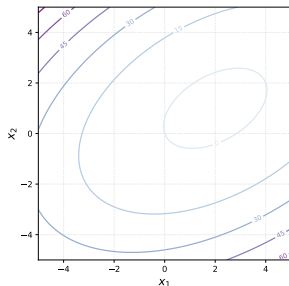
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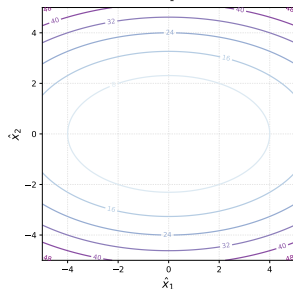
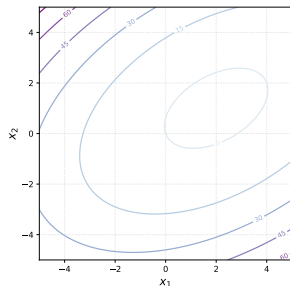
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## Convergence analysis

Now we can work with the function  $f(x) = \frac{1}{2}x^T \Lambda x$  with  $x^* = 0$  without loss of generality (drop the hat from the  $\hat{x}$ )

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Let's use constant stepsize  $\alpha^k = \alpha$ . Convergence condition:

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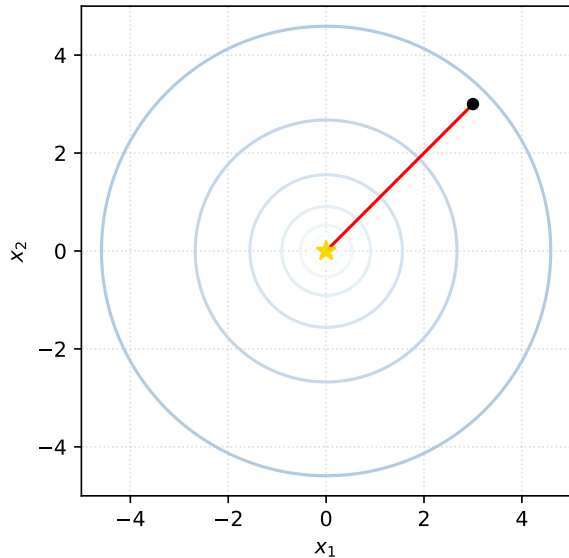
## Convergence analysis

So, we have a linear convergence in the domain with rate  $\frac{\kappa-1}{\kappa+1} = 1 - \frac{2}{\kappa+1}$ , where  $\kappa = \frac{L}{\mu}$  is sometimes called *condition number* of the quadratic problem.

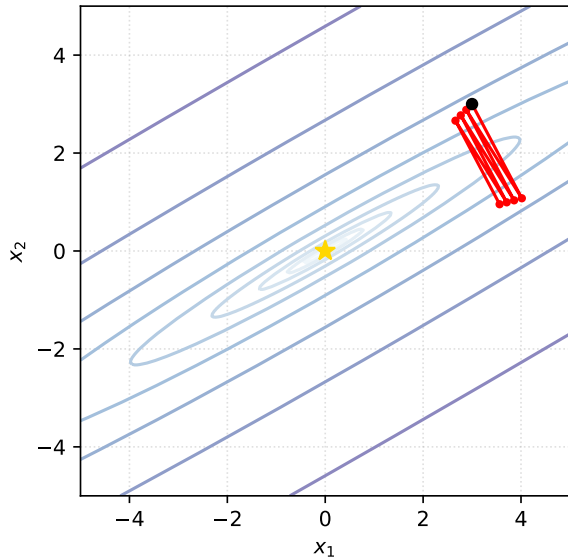
| $\kappa$ | $\rho$ | Iterations to decrease domain gap 10 times | Iterations to decrease function gap 10 times |
|----------|--------|--------------------------------------------|----------------------------------------------|
| 1.1      | 0.05   | 1                                          | 1                                            |
| 2        | 0.33   | 3                                          | 2                                            |
| 5        | 0.67   | 6                                          | 3                                            |
| 10       | 0.82   | 12                                         | 6                                            |
| 50       | 0.96   | 58                                         | 29                                           |
| 100      | 0.98   | 116                                        | 58                                           |
| 500      | 0.996  | 576                                        | 288                                          |
| 1000     | 0.998  | 1152                                       | 576                                          |

## Condition number $\kappa$

$\kappa = 1.0$



$\kappa = 100.0$



## Polyak-Lojasiewicz smooth case

## Polyak-Lojasiewicz condition. Linear convergence of gradient descent without convexity

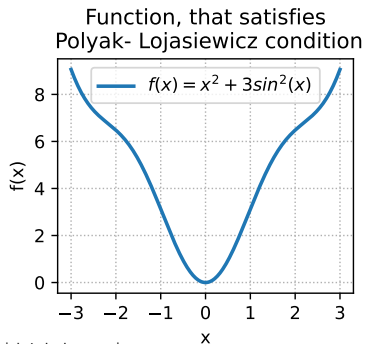
PL inequality holds if the following condition is satisfied for some  $\mu > 0$ ,

$$\|\nabla f(x)\|^2 \geq 2\mu(f(x) - f^*) \quad \forall x$$

It is interesting, that the Gradient Descent algorithm might converge linearly even without convexity.

The following functions satisfy the PL condition but are not convex. [🔗Link to the code](#)

$$f(x) = x^2 + 3\sin^2(x)$$



## Polyak-Lojasiewicz condition. Linear convergence of gradient descent without convexity

PL inequality holds if the following condition is satisfied for some  $\mu > 0$ ,

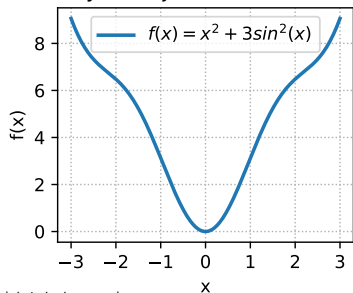
$$\|\nabla f(x)\|^2 \geq 2\mu(f(x) - f^*) \quad \forall x$$

It is interesting, that the Gradient Descent algorithm might converge linearly even without convexity.

The following functions satisfy the PL condition but are not convex. [🔗Link to the code](#)

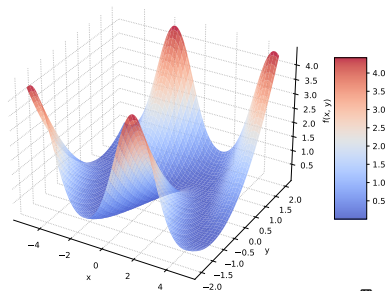
$$f(x) = x^2 + 3\sin^2(x)$$

Function, that satisfies  
Polyak-Lojasiewicz condition



$$f(x, y) = \frac{(y - \sin x)^2}{2}$$

Non-convex PL function



# Convergence analysis

## i Theorem

Consider the Problem

$$f(x) \rightarrow \min_{x \in \mathbb{R}^d}$$

and assume that  $f$  is  $\mu$ -Polyak-Lojasiewicz and  $L$ -smooth, for some  $L \geq \mu > 0$ .

Consider  $(x^k)_{k \in \mathbb{N}}$  a sequence generated by the gradient descent constant stepsize algorithm, with a stepsize satisfying  $0 < \alpha \leq \frac{1}{L}$ . Then:

$$f(x^k) - f^* \leq (1 - \alpha\mu)^k (f(x^0) - f^*).$$



## Convergence analysis

We can use  $L$ -smoothness, together with the update rule of the algorithm, to write

$$f(x^{k+1}) \leq f(x^k) + \langle \nabla f(x^k), x^{k+1} - x^k \rangle + \frac{L}{2} \|x^{k+1} - x^k\|^2$$

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where in the last inequality we used our hypothesis on the stepsize that  $\alpha L \leq 1$ .

We can now use the Polyak-Lojasiewicz property to write:

$$f(x^{k+1}) \leq f(x^k) - \alpha\mu(f(x^k) - f^*).$$

The conclusion follows after subtracting  $f^*$  on both sides of this inequality and using recursion.

## Any $\mu$ -strongly convex differentiable function is a PL-function

### Theorem

If a function  $f(x)$  is differentiable and  $\mu$ -strongly convex, then it is a PL function.

### Proof

By first order strong convexity criterion:

$$f(y) \geq f(x) + \nabla f(x)^T(y - x) + \frac{\mu}{2}\|y - x\|_2^2$$

Putting  $y = x^*$ :

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$$\begin{aligned} \text{Let } a &= \frac{1}{\sqrt{\mu}} \nabla f(x) \text{ and} \\ b &= \sqrt{\mu}(x - x^*) - \frac{1}{\sqrt{\mu}} \nabla f(x) \end{aligned}$$

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Let  $a = \frac{1}{\sqrt{\mu}} \nabla f(x)$  and

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$$f(x) - f(x^*) \leq \frac{1}{2} \left( \frac{1}{\mu} \|\nabla f(x)\|_2^2 - \left\| \sqrt{\mu}(x - x^*) - \frac{1}{\sqrt{\mu}} \nabla f(x) \right\|_2^2 \right)$$

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$$f(x) - f(x^*) \leq \frac{1}{2\mu} \|\nabla f(x)\|_2^2,$$

which is exactly the PL condition. It means, that we already have linear convergence proof for any strongly convex function.

## Smooth convex case

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Consider the Problem

$$f(x) \rightarrow \min_{x \in \mathbb{R}^d}$$

and assume that  $f$  is convex and  $L$ -smooth, for some  $L > 0$ .

Let  $(x^k)_{k \in \mathbb{N}}$  be the sequence of iterates generated by the gradient descent constant stepsize algorithm, with a stepsize satisfying  $0 < \alpha \leq \frac{1}{L}$ . Then, for all  $x^* \in \operatorname{argmin} f$ , for all  $k \in \mathbb{N}$  we have that

$$f(x^k) - f^* \leq \frac{\|x^0 - x^*\|^2}{2\alpha k}.$$

## Convergence analysis

- As it was before, we first use smoothness:

$$\begin{aligned}f(x^{k+1}) &\leq f(x^k) + \langle \nabla f(x^k), x^{k+1} - x^k \rangle + \frac{L}{2} \|x^{k+1} - x^k\|^2 \\&= f(x^k) - \alpha \|\nabla f(x^k)\|^2 + \frac{L\alpha^2}{2} \|\nabla f(x^k)\|^2 \\&= f(x^k) - \frac{\alpha}{2} (2 - L\alpha) \|\nabla f(x^k)\|^2 \\&\leq f(x^k) - \frac{\alpha}{2} \|\nabla f(x^k)\|^2, \\f(x^k) - f(x^{k+1}) &\geq \frac{1}{2L} \|\nabla f(x^k)\|^2 \text{ if } \alpha = \frac{1}{L}\end{aligned}\tag{1}$$

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Let  $a = x^k - x^*$  and  $b = x^k - x^* - \alpha \nabla f(x^k)$ . Then  $a + b = \alpha \nabla f(x^k)$  and  $a - b = 2(x^k - x^* - \frac{\alpha}{2} \nabla f(x^k))$ .

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(3)

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## Convergence analysis

- Due to the monotonic decrease at each iteration  $f(x^{i+1}) < f(x^i)$ :

$$kf(x^k) \leq \sum_{i=0}^{k-1} f(x^{i+1})$$

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$$2\alpha k f(x^k) - 2\alpha k f^* \leq 2\alpha \sum_{i=0}^{k-1} (f(x^{i+1}) - f^*) \leq \|x^0 - x^*\|_2^2$$
$$f(x^k) - f^* \leq \frac{\|x^0 - x^*\|_2^2}{2\alpha k}$$

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$$f(x^k) - f^* \leq \frac{\|x^0 - x^*\|_2^2}{2\alpha k} \leq \frac{L\|x^0 - x^*\|_2^2}{2k}$$

# Summary

Gradient Descent:

$$\min_{x \in \mathbb{R}^n} f(x)$$

$$x^{k+1} = x^k - \alpha^k \nabla f(x^k)$$

smooth (non-convex)

$$\|\nabla f(x^k)\|^2 \sim \mathcal{O}\left(\frac{1}{k}\right)$$

$$k_\varepsilon \sim \mathcal{O}\left(\frac{1}{\varepsilon}\right)$$

smooth & convex

$$f(x^k) - f^* \sim \mathcal{O}\left(\frac{1}{k}\right)$$

$$k_\varepsilon \sim \mathcal{O}\left(\frac{1}{\varepsilon}\right)$$

smooth & strongly convex (or PL)

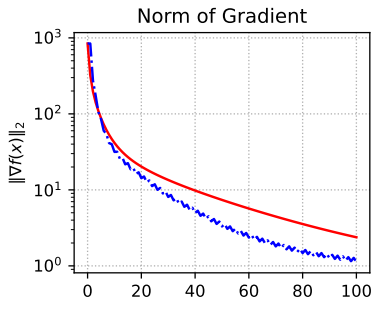
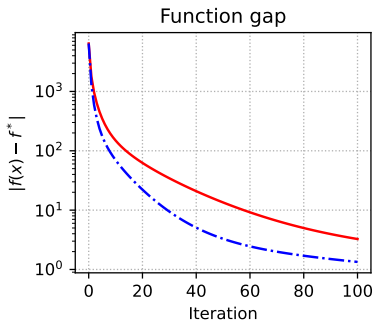
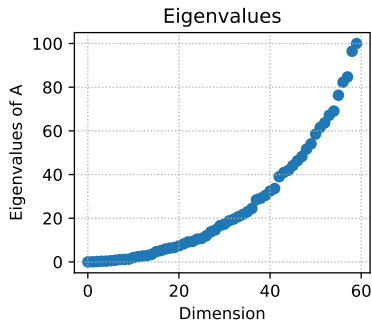
$$\|x^k - x^*\|^2 \sim \mathcal{O}\left(\left(1 - \frac{\mu}{L}\right)^k\right)$$

$$k_\varepsilon \sim \mathcal{O}\left(n \log \frac{1}{\varepsilon}\right)$$

# Numerical experiments

$$f(x) = \frac{1}{2}x^T A x - b^T x \rightarrow \min_{x \in \mathbb{R}^n}$$

Convex quadratics.  $n=60$ , random matrix.



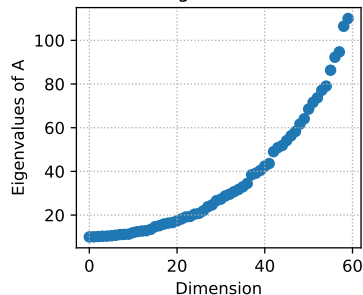
— Gradient Descent    - - - Steepest Descent

# Numerical experiments

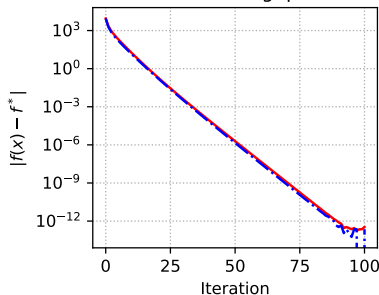
$$f(x) = \frac{1}{2}x^T A x - b^T x \rightarrow \min_{x \in \mathbb{R}^n}$$

Strongly convex quadratics.  $n=60$ , random matrix.

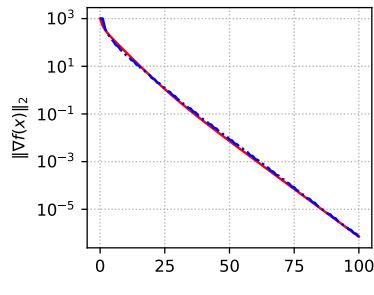
Eigenvalues



Function gap



Norm of Gradient

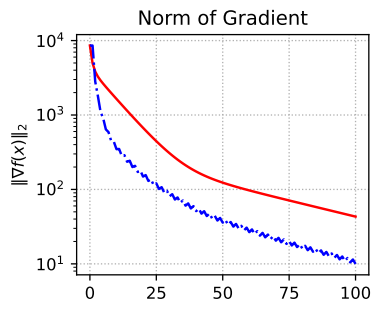
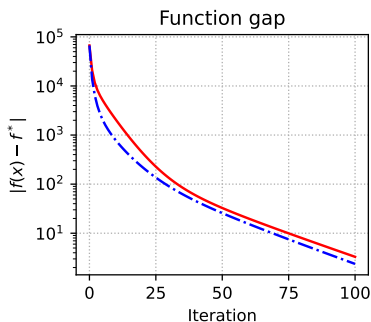
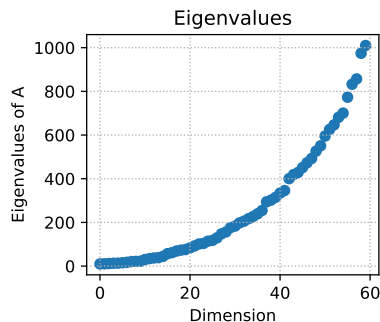


— Gradient Descent    - - - Steepest Descent

# Numerical experiments

$$f(x) = \frac{1}{2}x^T Ax - b^T x \rightarrow \min_{x \in \mathbb{R}^n}$$

Strongly convex quadratics.  $n=60$ , random matrix.

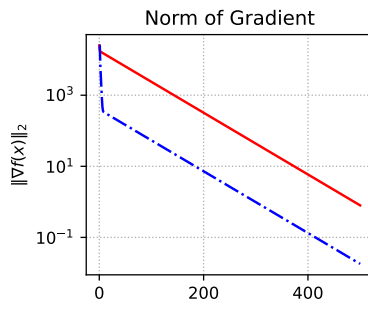
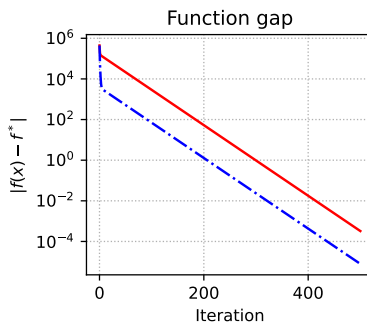
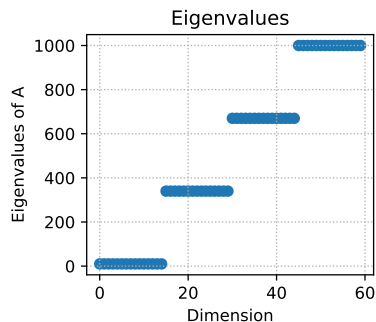


— Gradient Descent    - - - Steepest Descent

# Numerical experiments

$$f(x) = \frac{1}{2}x^T A x - b^T x \rightarrow \min_{x \in \mathbb{R}^n}$$

Strongly convex quadratics.  $n=60$ , clustered matrix.

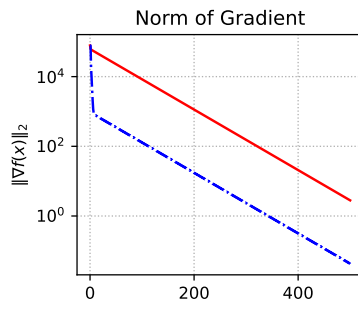
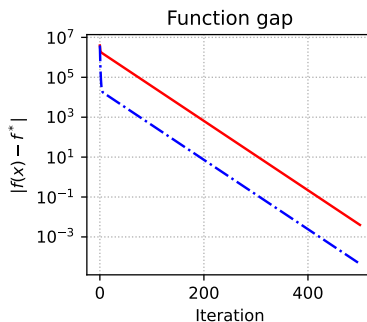
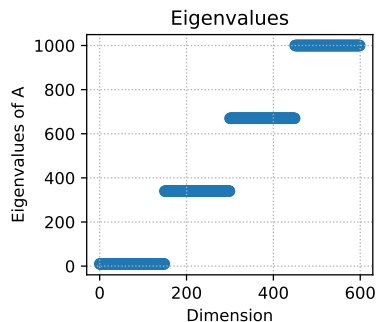


— Gradient Descent    -.- Steepest Descent

# Numerical experiments

$$f(x) = \frac{1}{2}x^T A x - b^T x \rightarrow \min_{x \in \mathbb{R}^n}$$

Strongly convex quadratics.  $n=600$ , clustered matrix.



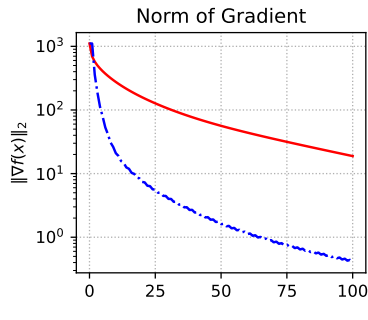
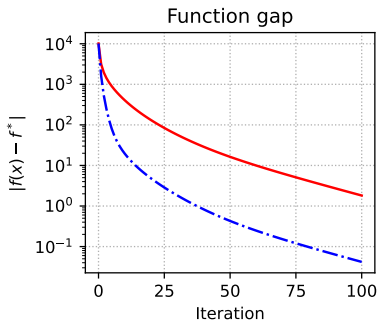
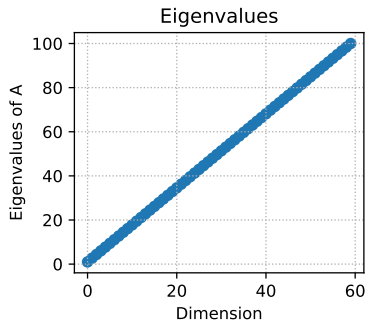
— Gradient Descent    -.- Steepest Descent



# Numerical experiments

$$f(x) = \frac{1}{2}x^T A x - b^T x \rightarrow \min_{x \in \mathbb{R}^n}$$

Strongly convex quadratics.  $n=60$ , uniform spectrum matrix.

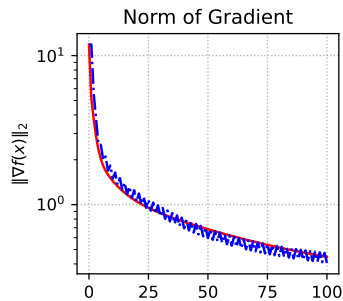
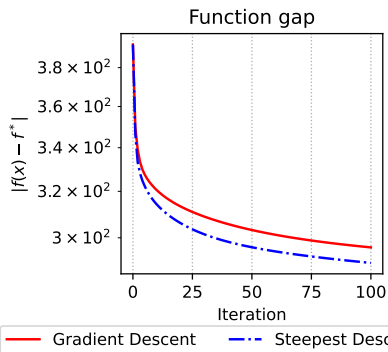
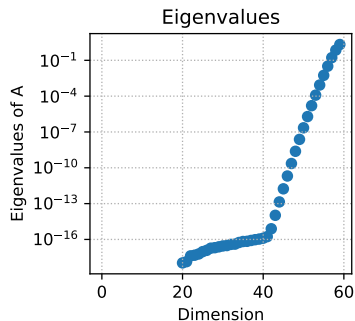


— Gradient Descent    -.- Steepest Descent

# Numerical experiments

$$f(x) = \frac{1}{2}x^T A x - b^T x \rightarrow \min_{x \in \mathbb{R}^n}$$

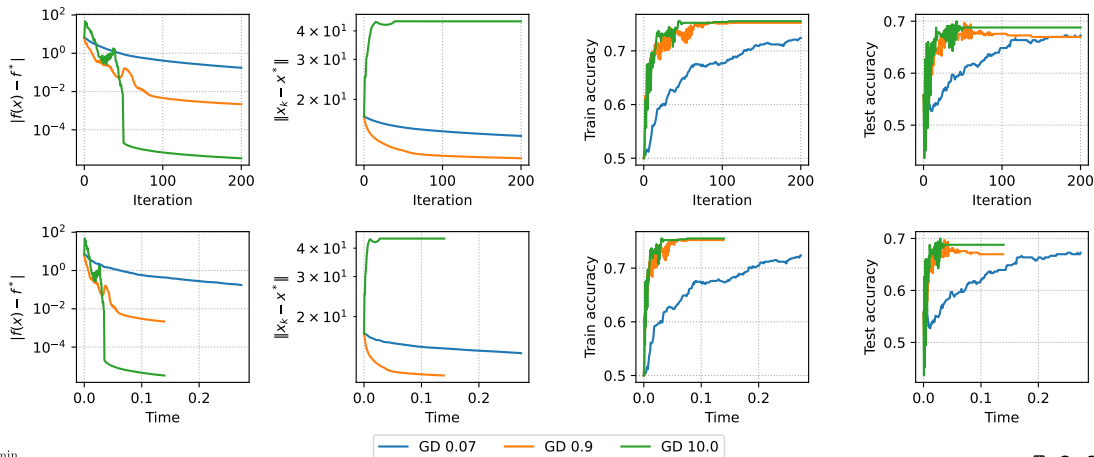
Strongly convex quadratics.  $n=60$ , Hilbert matrix.



# Numerical experiments

$$f(x) = \frac{\mu}{2} \|x\|_2^2 + \frac{1}{m} \sum_{i=1}^m \log(1 + \exp(-y_i \langle a_i, x \rangle)) \rightarrow \min_{x \in \mathbb{R}^n}$$

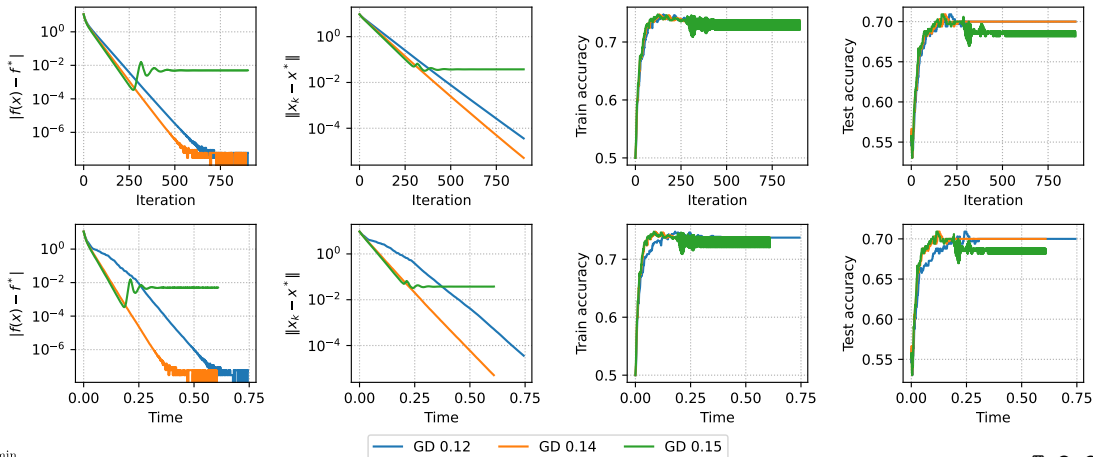
Convex binary logistic regression.  $\mu=0$ .



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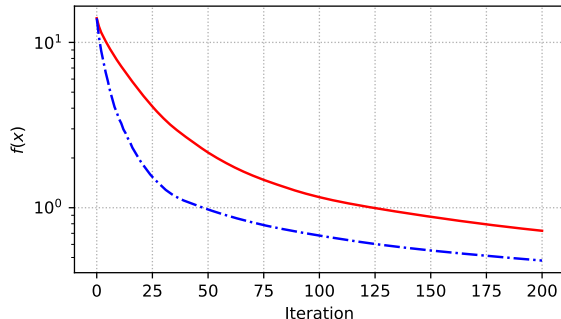
Strongly convex binary logistic regression.  $\mu=0.1$ .



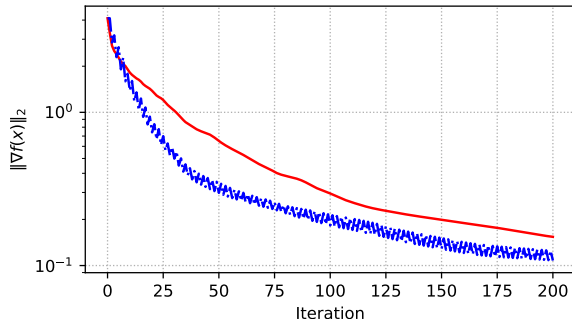
## Numerical experiments

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Regularized binary logistic regression.  $n=300$ .  $m=1000$ .  $\mu=0$



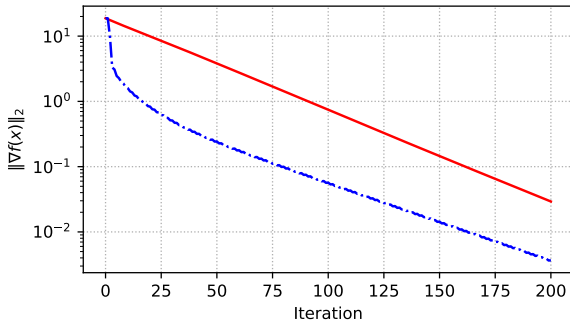
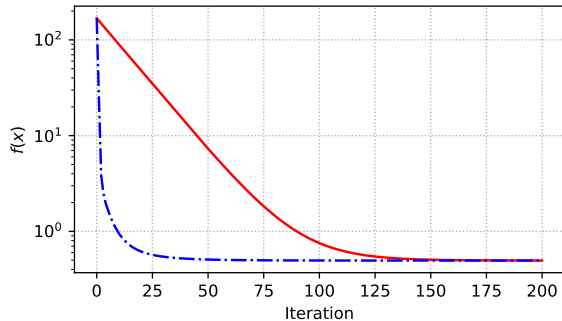
— Gradient Descent    -.- Steepest Descent



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Regularized binary logistic regression.  $n=300$ .  $m=1000$ .  $\mu=1$



— Gradient Descent    -.- Steepest Descent